

Implementation / Feasibility Plan

The numbers that show ASTRA works, and how it goes from a working model to state-wide use

ASTRA runs on ordinary hardware with large safety margins.

Every number below shows its working, so anyone can check it. Assumptions are stated up front, and all of them are deliberately on the heavy side.

Feasibility at a glance

What it delivers

< 1 s

to save every answer on the PC **and** both centre servers (control room copy within a minute)

< 90 s

to continue on any PC with the same answers, question order and screen position

< 30 s

to send candidates an official update after a failure, in Hindi and English

± 2 s

time precision: lost exam time is measured and returned to within 2 seconds

What it takes to run

99.6%

of a 100 Mbps centre network left free at peak (only 0.4% used)

0.1%

of a 10 Mbps internet line used to sync with the control room

≈ ₹5

hardware cost per candidate per exam, over 5 years of use

1 in 10,000

chance that both centre servers fail in the same exam

Assumptions used for every calculation

A typical centre has **300 candidates** taking a **3-hour exam** with **150 questions**. Each candidate makes **300 saves** (answers, changes, marking for review). Each saved change is **150 bytes** after encryption and signing. Each PC sends a **200-byte health signal every 2 seconds**. The centre network is **100 Mbps** and the internet line is **10 Mbps**.

1. Data size and network load (one centre)

What we calculate	Working	Result
Size of one candidate's full backup (Capsule)	150 questions × 6 bytes + 150 bytes of security data	≈ 1 KB (up to 3 KB with extras)
Data saved per centre per exam	300 candidates × 300 saves × 150 bytes	13.5 MB (about 3 phone photos)
Peak answer traffic	300 candidates saving once every 5 s × 150 bytes × 2 servers (main + spare)	144 kbps
Health-signal traffic	300 PCs ÷ 2 s × 200 bytes	240 kbps
Total peak load on the centre network	144 + 240 = 384 kbps out of 100,000 kbps	0.4% used → 260× headroom
Average internet use (sync to control room)	13.5 MB ÷ 10,800 s	10 kbps = 0.1% of a 10 Mbps line

What we calculate	Working	Result
Centre server database writes needed	$300 \div 5 \text{ s}$	60 writes/s , vs 1,000+/s typical for SQLite on an SSD (~ 16× headroom)

In plain words: the backup traffic is so small that even a slow 4G connection can carry a whole centre's data.

2. State-wide scale (one exam shift)

Example: **5,00,000 candidates** in one shift across **1,667 centres** of 300 candidates each.

What we calculate	Working	Result
Saved changes reaching the control room	$5,00,000 \times 300$	15 crore records per shift
Average records per second	$15 \text{ crore} \div 10,800 \text{ s}$	$\approx$ 13,900 records/s , vs 1,00,000+/s typical PostgreSQL bulk loading (~ 7× headroom)
Data per shift	$15 \text{ crore} \times 150 \text{ bytes}$	22.5 GB (fits on any modern server)
Control-room internet needed	$22.5 \text{ GB} \div 10,800 \text{ s}$	$\approx$ 17 Mbps
Upload requests	$1,667 \text{ centres} \times 1 \text{ batch per minute}$	$\approx$ 28 requests/s

Most important: the control room is **not needed to keep exams running**. If it slows down or fails, centres keep going and send their data later.

3. Recovery speed (time budget after a power cut)

Stage	Working	Time
Notice the problem	PC misses 3 health signals ($3 \times 2 \text{ s}$) + 4 s to confirm the whole room is affected	$\approx$ 10 s
Identify it	Fast AI and rules decide in milliseconds	< 1 s
Tell candidates	Ready-made message sent after the problem is identified	< 30 s after the failure
Continue the exam	PC restart $\approx 40 \text{ s}$ + log-in and invigilator approval $\approx 30 \text{ s}$ + loading a 3 KB backup over the network ($< 0.1 \text{ s}$)	$\approx$ 70 s (target < 90 s)
Answers lost	Each answer is written to disk before "saved" appears	0 confirmed answers
Time wrongly charged	The clock stops at the last health signal, which is at most 2 s before the failure	$\leq$ 2 s, always in the candidate's favour

4. Reliability

Illustrative failure rates; the method matters more than the exact inputs.

What we calculate	Working	Result
Centre stops because its server failed	One server failing in 1 of 100 exams ($p = 0.01$). Two servers on separate power must both fail: 0.01×0.01	1 in 10,000 exams (100× safer than one server)
A confirmed answer is lost	All 3 copies must be lost together: PC disk (0.001) $\times$ both centre servers (0.0001) $\times$ control room (0.001)	$\approx$ 1 in 10 billion

What we calculate	Working	Result
Exam stops because the internet or control room failed	The exam doesn't depend on them	0

5. AI computing load

AI task	Working	Result
Failure prediction (LightGBM)	1 prediction per room every 10 s; each takes about a millisecond on a normal CPU	< 1% of one CPU core
Reading invigilator reports (Laya)	0.2–0.5 s per question on a CPU (as reported by its makers) × a few reports per hour	Negligible
Writing explanations (local language model)	About 150 words at 5–15 words/s on a laptop CPU	10–30 s, off the critical path; a template appears instantly
Training the prediction model	10,000 simulated exam-hours × 360 readings per hour = 36 lakh rows	Minutes on a laptop

6. Cost (indicative Indian prices, to be confirmed in Phase 1)

Item per centre	Working	Cost
2 centre servers (business desktops)	2 × ₹45,000	₹90,000
2 battery backups (1 kVA online UPS)	2 × ₹18,000	₹36,000
Network switch + 4G backup router	₹15,000 + ₹5,000	₹20,000
One-time total		≈ ₹1.46 lakh
Software licences	All open-source	₹0
Cost per exam day	₹1.46 lakh ÷ (20 exam days a year × 5 years)	≈ ₹1,460
Cost per candidate per exam	₹1,460 ÷ 300 candidates	≈ ₹5

One avoided cancellation pays for the hardware of 6 centres.

A cancelled shift at a 300-candidate centre costs candidates about $300 \times ₹3,100$ (travel, stay, a lost day) = ₹9.3 lakh. $₹9.3 \text{ lakh} \div ₹1.46 \text{ lakh} \approx 6.4$ centres. This doesn't include the exam body's own cost of holding the exam again.

7. Step-by-step roll-out

Phase	Time	What happens	Move on when (measured)
0 • Working model	1–2 months	Tested on simulated centres with every failure type	0 answers lost in 100% of test failures; resume in under 90 s
1 • Watch only	2–3 months	3–5 real centres, only watching	Live view matches reality; AI warnings checked against real incidents
2 • Assist	3 months	Instructions, messages, decision help switched on	First message in under 30 s; records match the current system 100%

Phase	Time	What happens	Move on when (measured)
3 • Full trial	3 months	One complete exam uses ASTRA for backup and recovery	0 lost answers; 0 needless re-exams
4 • State-wide	6–12 months	All centres of selected exam bodies	Fewer cancellations than the past 3 years' average

8. What each centre needs, and people

 Current exam PCs Locked-down exam setup, or just a monitoring program in early phases	 2 small servers + 2 UPS Main and spare, on different power lines	 Network + CCTV Separate exam and camera networks, existing cameras, 4G backup
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- **Training:** invigilators 1–2 hours; centre staff 1 day; officials 1 day. A **backup drill before every exam** doubles as practice.
- **Set by the exam authority:** maximum pause time, how lost time is returned, who approves each decision, and CCTV retention.
- **Compliance:** Data Protection Act 2023, Public Examinations Act 2024, and an independent CERT-In empanelled security audit before Phase 3.

9. Risks and how we handle them

Risk	What we do
Centres with poor power backup	The health check exposes it before the exam; the centre server's own UPS protects answers
Exam companies unwilling to connect	Start in watch-only mode; standard connection format; add to future contracts
AI mistakes early on	Officials approve important actions; rules as backup; retraining on real data
Staff misusing access	"Blind courier" server, two-person approval, every action recorded
Very large exams	Centres run independently; the control room has 7–16× headroom (sections 1–2)